Quantum Computing: Phase 3
Density Matrix, Purification & Decomposition

Cai Yundi Elliot

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1 Density Matrix Formalism

1.1 Motivation: Why Density Matrices?

So far, we've described quantum states using state vectors $|\psi\rangle$. This works for **pure states** – states we know perfectly.

But what if:

- We don't know the exact state (incomplete knowledge)
- The system is entangled with another system we don't have access to
- There's noise or decoherence in the system

In these cases, we need **density matrices** to describe **mixed states**.

1.2 Physical Significance: What Does the Density Matrix Represent?

Key Insight: The density matrix ρ encodes both quantum superposition AND classical uncertainty.

Physical Meaning:

- **Diagonal elements** (ρ_{ii}): Population probabilities (classical)
- **Off-diagonal elements** ($\rho_{ij}, i \neq j$): Coherences (quantum interference)
- **Pure state**: Maximum coherence, $\mathrm{Tr}(\rho^2) = 1$
- **Mixed state**: Reduced coherence, $\mathrm{Tr}(\rho^2) < 1$

Example: Compare pure state $|+\rangle$ vs mixed state (50% $|0\rangle$, 50% $|1\rangle$):

$$\rho_+ = \frac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \quad \text{vs} \quad \rho_{\text{mix}} = \frac{1}{2}\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

Physical interpretation:

- Both have same diagonal (50% probability to measure $|0\rangle$ or $|1\rangle$)
- ρ_+ has off-diagonal coherence (can interfere, shows wave behavior)
- ρ_{mix} has no coherence (classical mixture, no interference)

Real-world significance:

- **Quantum computing**: Decoherence destroys off-diagonal elements (quantum $\to$ classical)
- **Quantum measurement**: Measurement collapses coherences
- **Thermal states**: Mixed states describe systems at finite temperature
- **Entanglement**: Reduced density matrices of entangled systems are mixed

1.3 Definition

Definition 1.1 (Density Matrix)**.** *For an ensemble $\{p_i, |\psi_i\rangle\}$ where p_i is the probability of state $|\psi_i\rangle$, the density matrix is:*

$$\rho = \sum_i p_i |\psi_i\rangle\langle\psi_i| \tag{1}$$

where $\sum_i p_i = 1$ and $p_i \geq 0$.

Physical Interpretation:
The density matrix unifies two types of uncertainty:

- **Quantum uncertainty**: Intrinsic to each state $|\psi_i\rangle$ (superposition)
- **Classical uncertainty**: Which state $|\psi_i\rangle$ the system is actually in (probability p_i)

For a pure state, there's only one term in the sum ($p_1 = 1$), so $\rho = |\psi\rangle\langle\psi|$. All uncertainty is quantum. For a mixed state, multiple terms contribute, and we have both quantum and classical uncertainty.

1.4 Pure States as Density Matrices: Derivation

Claim: For a pure state $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, the density matrix is:

$$\rho = |\psi\rangle\langle\psi| = \begin{pmatrix} |\alpha|^2 & \alpha\beta^* \\ \alpha^*\beta & |\beta|^2 \end{pmatrix} \tag{2}$$

Proof:
Step 1: Write the bra vector:

$$\langle\psi| = (\alpha|0\rangle + \beta|1\rangle)^\dagger = \alpha^*\langle 0| + \beta^*\langle 1| \tag{3}$$

Step 2: Compute the outer product:

$$\rho = |\psi\rangle\langle\psi| = (\alpha|0\rangle + \beta|1\rangle)(\alpha^*\langle 0| + \beta^*\langle 1|) \tag{4}$$

Step 3: Expand using distributivity:

$$\rho = \alpha\alpha^*|0\rangle\langle 0| + \alpha\beta^*|0\rangle\langle 1| + \beta\alpha^*|1\rangle\langle 0| + \beta\beta^*|1\rangle\langle 1| \tag{5}$$

Step 4: Use $|i\rangle\langle j|$ as basis matrices:

$$|0\rangle\langle 0| = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad |0\rangle\langle 1| = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \tag{6}$$

$$|1\rangle\langle 0| = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \quad |1\rangle\langle 1| = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \tag{7}$$

Step 5: Substitute and simplify:

$$\begin{aligned} \rho &= |\alpha|^2 \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + \alpha\beta^* \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} + \alpha^*\beta \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} + |\beta|^2 \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \\ &= \begin{pmatrix} |\alpha|^2 & \alpha\beta^* \\ \alpha^*\beta & |\beta|^2 \end{pmatrix} \end{aligned} \tag{8}$$

Example: For $|+\rangle = \frac{|0\rangle+|1\rangle}{\sqrt{2}}$ (where $\alpha = \beta = \frac{1}{\sqrt{2}}$):

$$\rho_+ = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{pmatrix} = \frac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \tag{9}$$

Physical Significance:

The diagonal elements $|\alpha|^2$ and $|\beta|^2$ are the measurement probabilities in the computational basis. The off-diagonal elements $\alpha\beta^*$ and $\alpha^*\beta$ are called **coherences**—they encode the relative phase between $|0\rangle$ and $|1\rangle$, which enables interference. When these coherences vanish (due to decoherence or measurement), the quantum state becomes a classical mixture.

1.5 Code: Create Density Matrix (Level 21)

File: `code/density_matrix/21_create_density_matrix.py`

```python
import numpy as np

def density_matrix(psi):
    """Create density matrix from state vector: rho = |psi><psi|"""
    return np.outer(psi, psi.conj())

# Pure state |+>
psi_plus = np.array([1/np.sqrt(2), 1/np.sqrt(2)])
rho_plus = density_matrix(psi_plus)
print("Density matrix of |+>:")
print(rho_plus)
# Output: [[0.5 0.5]
#          [0.5 0.5]]

# Verify properties
print("Hermitian:", np.allclose(rho_plus, rho_plus.conj().T))
print("Trace:", np.trace(rho_plus))
print("Purity (Tr(rho^2)):", np.trace(rho_plus @ rho_plus).real)
```

Listing 1: Create density matrix from state vector (Level 21)

1.6 Mixed States: Detailed Example

Example: 50% $|0\rangle$, 50% $|1\rangle$ (classical mixture):

Step 1: Write the ensemble:

$$\{p_0 = 0.5, |\psi_0\rangle = |0\rangle\}, \quad \{p_1 = 0.5, |\psi_1\rangle = |1\rangle\} \tag{10}$$

Step 2: Apply definition:

$$\rho_{\text{mix}} = 0.5|0\rangle\langle 0| + 0.5|1\rangle\langle 1| \tag{11}$$

Step 3: Compute each term:

$$|0\rangle\langle 0| = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad |1\rangle\langle 1| = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \tag{12}$$

Step 4: Add:

$$\rho_{\text{mix}} = 0.5\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + 0.5\begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} = \frac{1}{2}\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \tag{13}$$

Physical Significance:

This represents **classical ignorance**, not quantum superposition:

- The system IS either $|0\rangle$ or $|1\rangle$ (we just don't know which)
- No interference pattern possible (unlike $|+\rangle$)
- Equivalent to flipping a classical coin and preparing $|0\rangle$ or $|1\rangle$
- Called "incoherent mixture" or "statistical mixture"

Key difference: Compare with pure state $|+\rangle$:

$$\rho_+ = \frac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \quad \text{vs} \quad \rho_{\text{mix}} = \frac{1}{2}\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \tag{14}$$

Same diagonal (measurement probabilities in computational basis), but different off-diagonal elements (coherences).

Why Coherences Matter:

The off-diagonal elements (coherences) are what make quantum mechanics *quantum*. They enable interference effects that classical probability cannot reproduce. When a quantum system interacts with its environment (decoherence), these coherences decay exponentially, and the density matrix becomes diagonal—indistinguishable from a classical mixture. This is why maintaining coherence is the central challenge in building quantum computers.

1.7 Code: Mixed State Ensemble (Level 22)

File: `code/density_matrix/22_mixed_state_ensemble.py`

```python
def mixed_state_ensemble(states, probs):
    """Create density matrix from ensemble {p_i, |psi_i>}"""
    rho = np.zeros((2, 2), dtype=complex)
    for p, psi in zip(probs, states):
        rho += p * np.outer(psi, psi.conj())
    return rho

# 50% |0>, 50% |1> (classical mixture)
states = [np.array([1, 0]), np.array([0, 1])]
probs = [0.5, 0.5]
rho_mix = mixed_state_ensemble(states, probs)
print("Mixed state density matrix:")
print(rho_mix)
# Output: [[0.5 0. ]
#          [0.  0.5]]

# Compare with pure state |+>
psi_plus = np.array([1/np.sqrt(2), 1/np.sqrt(2)])
rho_plus = density_matrix(psi_plus)
print("\nPure state |+> density matrix:")
print(rho_plus)

print("\nKey difference:")
print("- Same diagonal (measurement probabilities)")
print("- Different off-diagonal (coherences)")
```

Listing 2: Create mixed state from ensemble (Level 22)

1.8 Properties of Density Matrices: Complete Proofs

Theorem 1.1 (Density Matrix Properties)**.** *A valid density matrix ρ must satisfy:*

1. ***Hermitian:*** $\rho^\dagger = \rho$
2. ***Unit trace:*** $Tr(\rho) = 1$
3. ***Positive semi-definite:*** $\langle\psi|\rho|\psi\rangle \geq 0$ *for all* $|\psi\rangle$

Physical Significance of Properties:

- **Hermitian**: Ensures measurement outcomes are real numbers (observable physics)
- **Unit trace**: Total probability = 1 (probability conservation)
- **Positive semi-definite**: All probabilities are non-negative (physical requirement)

Proof of Property 1 (Hermitian):
Step 1: Start with definition:

$$\rho = \sum_i p_i |\psi_i\rangle\langle\psi_i| \tag{15}$$

Step 2: Take Hermitian conjugate:

$$\rho^\dagger = \left(\sum_i p_i |\psi_i\rangle\langle\psi_i|\right)^\dagger = \sum_i p_i^* (|\psi_i\rangle\langle\psi_i|)^\dagger \tag{16}$$

Step 3: Use $(|a\rangle\langle b|)^\dagger = |b\rangle\langle a|$ and p_i is real:

$$\rho^\dagger = \sum_i p_i (|\psi_i\rangle\langle\psi_i|)^\dagger = \sum_i p_i |\psi_i\rangle\langle\psi_i| \tag{17}$$

Step 4: Conclude:

$$\rho^\dagger = \rho \tag{18}$$

Proof of Property 2 (Unit Trace):
Step 1: Use linearity of trace:

$$\mathrm{Tr}(\rho) = \mathrm{Tr}\left(\sum_i p_i |\psi_i\rangle\langle\psi_i|\right) = \sum_i p_i \mathrm{Tr}(|\psi_i\rangle\langle\psi_i|) \tag{19}$$

Step 2: Use cyclic property $\mathrm{Tr}(|a\rangle\langle b|) = \langle b|a\rangle$:

$$\mathrm{Tr}(|\psi_i\rangle\langle\psi_i|) = \langle\psi_i|\psi_i\rangle = 1 \quad \text{(normalized)} \tag{20}$$

Step 3: Sum probabilities:

$$\mathrm{Tr}(\rho) = \sum_i p_i \cdot 1 = \sum_i p_i = 1 \tag{21}$$

Proof of Property 3 (Positive Semi-Definite):

Step 1: Consider arbitrary state $|\phi\rangle$:

$$\langle\phi|\rho|\phi\rangle = \langle\phi|\left(\sum_i p_i|\psi_i\rangle\langle\psi_i|\right)|\phi\rangle \tag{22}$$

Step 2: Distribute:

$$\langle\phi|\rho|\phi\rangle = \sum_i p_i\langle\phi|\psi_i\rangle\langle\psi_i|\phi\rangle \tag{23}$$

Step 3: Recognize $|\langle\phi|\psi_i\rangle|^2$:

$$\langle\phi|\rho|\phi\rangle = \sum_i p_i|\langle\phi|\psi_i\rangle|^2 \tag{24}$$

Step 4: Each term is non-negative ($p_i \geq 0$ and $|z|^2 \geq 0$):

$$\langle\phi|\rho|\phi\rangle \geq 0 \tag{25}$$

Physical Significance of Properties:

- **Hermitian**: Ensures measurement outcomes are real numbers (observable physics)
- **Unit trace**: Total probability = 1 (probability conservation)
- **Positive semi-definite**: All probabilities are non-negative (physical requirement)

Proof of Property 1 (Hermitian):
Step 1: Start with definition:

$$\rho = \sum_i p_i|\psi_i\rangle\langle\psi_i|$$

Step 2: Take Hermitian conjugate:

$$\rho^\dagger = \left(\sum_i p_i|\psi_i\rangle\langle\psi_i|\right)^\dagger = \sum_i p_i^*(|\psi_i\rangle\langle\psi_i|)^\dagger$$

Step 3: Use $(|a\rangle\langle b|)^\dagger = |b\rangle\langle a|$ and p_i is real:

$$\rho^\dagger = \sum_i p_i(|\psi_i\rangle\langle\psi_i|)^\dagger = \sum_i p_i|\psi_i\rangle\langle\psi_i|$$

Step 4: Conclude:

$$\rho^\dagger = \rho$$

Proof of Property 2 (Unit Trace):
Step 1: Use linearity of trace:

$$\mathrm{Tr}(\rho) = \mathrm{Tr}\left(\sum_i p_i|\psi_i\rangle\langle\psi_i|\right) = \sum_i p_i\mathrm{Tr}(|\psi_i\rangle\langle\psi_i|)$$

Step 2: Use cyclic property $\mathrm{Tr}(|a\rangle\langle b|) = \langle b|a\rangle$:

$$\mathrm{Tr}(|\psi_i\rangle\langle\psi_i|) = \langle\psi_i|\psi_i\rangle = 1 \quad \text{(normalized)}$$

Step 3: Sum probabilities:

$$\mathrm{Tr}(\rho) = \sum_i p_i \cdot 1 = \sum_i p_i = 1$$

Proof of Property 3 (Positive Semi-Definite):
Step 1: Consider arbitrary state $|\phi\rangle$:

$$\langle\phi|\rho|\phi\rangle = \langle\phi| \left(\sum_i p_i |\psi_i\rangle\langle\psi_i| \right) |\phi\rangle$$

Step 2: Distribute:

$$\langle\phi|\rho|\phi\rangle = \sum_i p_i \langle\phi|\psi_i\rangle\langle\psi_i|\phi\rangle$$

Step 3: Recognize $|\langle\phi|\psi_i\rangle|^2$:

$$\langle\phi|\rho|\phi\rangle = \sum_i p_i |\langle\phi|\psi_i\rangle|^2$$

Step 4: Each term is non-negative ($p_i \geq 0$ and $|z|^2 \geq 0$):

$$\langle\phi|\rho|\phi\rangle \geq 0$$

1.9 Code: Validate Density Matrix (Level 23)

File: `code/density_matrix/23_check_valid_density_matrix.py`

```python
def is_valid_density_matrix(rho, tol=1e-10):
    """Check if matrix is valid density matrix.

    Properties:
    1. Hermitian: rho^dagger = rho (real measurement outcomes)
    2. Unit trace: Tr(rho) = 1 (probability conservation)
    3. Positive semi-definite: all eigenvalues >= 0 (physical probabilities)
    """
    # Check Hermitian
    if not np.allclose(rho, rho.conj().T):
        return False, "Not Hermitian"

    # Check unit trace
    if not np.abs(np.trace(rho) - 1) < tol:
        return False, "Trace != 1"

    # Check positive semi-definite
    eigenvalues = np.linalg.eigvalsh(rho)
    if np.any(eigenvalues < -tol):
        return False, "Not positive semi-definite"

    return True, "Valid density matrix"

# Test valid density matrix
psi = np.array([1/np.sqrt(2), 1/np.sqrt(2)])
rho_valid = np.outer(psi, psi.conj())
valid, msg = is_valid_density_matrix(rho_valid)
print(f"|+> density matrix: {msg}")

# Test invalid matrix (non-Hermitian)
```

```python
rho_invalid = np.array([[0.5, 0.5], [0.3, 0.5]])
valid, msg = is_valid_density_matrix(rho_invalid)
print(f"Invalid matrix: {msg}")

# Test identity/2 (valid mixed state)
rho_max_mixed = np.eye(2) / 2
valid, msg = is_valid_density_matrix(rho_max_mixed)
print(f"Maximally mixed state: {msg}")
```

Listing 3: Verify density matrix properties (Level 23)

1.10 Measurement with Density Matrices: Derivation

Claim: For observable M with eigenvalues m_i and projectors P_i:

$$P(m_i) = \mathrm{Tr}(P_i \rho) \tag{26}$$

Physical Significance:
This formula unifies:

- **Quantum uncertainty**: Superposition within each $|\psi_k\rangle$
- **Classical uncertainty**: Ensemble average over p_k

Proof:
Step 1: For pure state $|\psi\rangle$, Born rule gives:

$$P(m_i) = \langle\psi|P_i|\psi\rangle \tag{27}$$

Step 2: For ensemble $\{p_k, |\psi_k\rangle\}$, average over ensemble:

$$P(m_i) = \sum_k p_k \langle\psi_k|P_i|\psi_k\rangle \tag{28}$$

Step 3: Use trace trick $\langle a|O|a\rangle = \mathrm{Tr}(O|a\rangle\langle a|)$:

$$P(m_i) = \sum_k p_k \mathrm{Tr}(P_i|\psi_k\rangle\langle\psi_k|) \tag{29}$$

Step 4: Use linearity of trace:

$$P(m_i) = \mathrm{Tr}\left(P_i \sum_k p_k |\psi_k\rangle\langle\psi_k|\right) \tag{30}$$

Step 5: Recognize density matrix:

$$P(m_i) = \mathrm{Tr}(P_i \rho) \tag{31}$$

Post-measurement state:

$$\rho' = \frac{P_i \rho P_i}{\mathrm{Tr}(P_i \rho)} \tag{32}$$

$$P(m_i) = \sum_k p_k \langle\psi_k|P_i|\psi_k\rangle$$

Step 3: Use trace trick $\langle a|O|a\rangle = \mathrm{Tr}(O|a\rangle\langle a|)$:

$$P(m_i) = \sum_k p_k \mathrm{Tr}(P_i|\psi_k\rangle\langle\psi_k|)$$

Step 4: Use linearity of trace:

$$P(m_i) = \mathrm{Tr}\left(P_i \sum_k p_k|\psi_k\rangle\langle\psi_k|\right)$$

Step 5: Recognize density matrix:

$$P(m_i) = \mathrm{Tr}(P_i\rho)$$

Post-measurement state:

$$\rho' = \frac{P_i\rho P_i}{\mathrm{Tr}(P_i\rho)}$$

1.11 Time Evolution: Derivation

Claim: For unitary U:

$$\rho' = U\rho U^\dagger \tag{33}$$

Physical Significance:
This is the **Liouville-von Neumann equation** in discrete form:

- Describes closed system evolution (no decoherence)
- Preserves purity: $\mathrm{Tr}((\rho')^2) = \mathrm{Tr}(\rho^2)$
- Preserves eigenvalues (information conservation)
- Reduces to Schrödinger equation for pure states

Proof:
Step 1: For pure state $|\psi\rangle$, time evolution is:

$$|\psi'\rangle = U|\psi\rangle \tag{34}$$

Step 2: Density matrix transforms as:

$$\rho' = |\psi'\rangle\langle\psi'| = (U|\psi\rangle)(U|\psi\rangle)^\dagger \tag{35}$$

Step 3: Use $(AB)^\dagger = B^\dagger A^\dagger$:

$$\rho' = U|\psi\rangle\langle\psi|U^\dagger \tag{36}$$

Step 4: For ensemble, apply to each term:

$$\rho' = \sum_i p_i U|\psi_i\rangle\langle\psi_i|U^\dagger = U\left(\sum_i p_i|\psi_i\rangle\langle\psi_i|\right)U^\dagger \tag{37}$$

Step 5: Conclude:

$$\rho' = U\rho U^\dagger \tag{38}$$

Why Unitary Evolution Matters:
Unitary evolution preserves the eigenvalues of ρ, which means the amount of uncertainty (or mixedness) stays constant in closed systems. This is the quantum version of Liouville's theorem in classical statistical mechanics—phase space volume is preserved. However, when a system interacts with an environment (open system), the evolution is no longer unitary, and purity can decrease. This is decoherence in action.

2 Bloch Sphere for Mixed States

2.1 Generalized Bloch Vector: Derivation

Claim: Any single-qubit density matrix can be written as:

$$\rho = \frac{1}{2}(I + \vec{r} \cdot \vec{\sigma})$$

where $\vec{r} = (r_x, r_y, r_z)$ is the Bloch vector and $\vec{\sigma} = (X, Y, Z)$.

Physical Significance:
The Bloch vector provides geometric intuition:

- **Direction**: Which "axis" the spin prefers
- **Length**: How "pure" the state is
- **Surface** ($|\vec{r}| = 1$): Pure states (maximum knowledge)
- **Interior** ($|\vec{r}| < 1$): Mixed states (partial knowledge)
- **Center** ($\vec{r} = 0$): Maximally mixed (minimum knowledge)

Proof: (See Section 1.5 for complete derivation)

2.2 Extracting Bloch Vector

From $\rho = \frac{1}{2}(I + r_x X + r_y Y + r_z Z)$:

$$r_x = \mathrm{Tr}(\rho X), \quad r_y = \mathrm{Tr}(\rho Y), \quad r_z = \mathrm{Tr}(\rho Z)$$

2.3 Code: Bloch Vector Extraction (Level 24)

File: `code/density_matrix/24_bloch_vector.py`

```python
def bloch_vector(rho):
    """Extract Bloch vector components from density matrix.

    Physical meaning:
    - Direction: spin preference axis
    - Length: purity (1=pure, 0=maximally mixed)
    """
    # Pauli matrices
    X = np.array([[0, 1], [1, 0]])
    Y = np.array([[0, -1j], [1j, 0]])
    Z = np.array([[1, 0], [0, -1]])

    # Bloch vector components
    rx = np.trace(rho @ X).real
    ry = np.trace(rho @ Y).real
    rz = np.trace(rho @ Z).real

    return np.array([rx, ry, rz])

# Pure state |+>
psi_plus = np.array([1/np.sqrt(2), 1/np.sqrt(2)])
rho_plus = np.outer(psi_plus, psi_plus.conj())
```

```python
print("Bloch vector of |+>:")
print(bloch_vector(rho_plus))
print(f"Length: {np.linalg.norm(bloch_vector(rho_plus)):.6f}")
# Expected: [1, 0, 0] (on x-axis), length = 1 (pure)

# Pure state |+i>
psi_i = np.array([1/np.sqrt(2), 1j/np.sqrt(2)])
rho_i = np.outer(psi_i, psi_i.conj())
print("\nBloch vector of |+i>:")
print(bloch_vector(rho_i))
print(f"Length: {np.linalg.norm(bloch_vector(rho_i)):.6f}")
# Expected: [0, 1, 0] (on y-axis), length = 1 (pure)

# Maximally mixed state
rho_mix = np.eye(2) / 2
print("\nBloch vector of maximally mixed state:")
print(bloch_vector(rho_mix))
print(f"Length: {np.linalg.norm(bloch_vector(rho_mix)):.6f}")
# Expected: [0, 0, 0] (at center), length = 0 (maximally mixed)
```

Listing 4: Extract Bloch vector from density matrix (Level 24)

2.4 Pure vs Mixed States: Proof

Claim:

- Pure states: $|\vec{r}| = 1$ (on the surface)
- Mixed states: $|\vec{r}| < 1$ (inside the sphere)
- Maximally mixed: $\vec{r} = 0$ (at the center, $\rho = I/2$)

Physical Significance:
The length $|\vec{r}|$ measures **how much we know** about the state:

- $|\vec{r}| = 1$: Complete knowledge (pure state)
- $0 < |\vec{r}| < 1$: Partial knowledge (mixed state)
- $|\vec{r}| = 0$: No knowledge (maximally mixed, infinite temperature)

Proof: (See Section 1.6 for complete derivation using purity $\mathrm{Tr}(\rho^2)$)

2.5 Examples Table

State	**Density Matrix**	**Bloch Vector**	**Type**
$\|0\rangle$	$\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$	$(0, 0, 1)$	Pure
$\|1\rangle$	$\begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$	$(0, 0, -1)$	Pure
$\|+\rangle$	$\frac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$	$(1, 0, 0)$	Pure
$\|+i\rangle$	$\frac{1}{2}\begin{pmatrix} 1 & -i \\ i & 1 \end{pmatrix}$	$(0, 1, 0)$	Pure
50% $\|0\rangle$, 50% $\|1\rangle$	$\frac{1}{2}\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$	$(0, 0, 0)$	Mixed

Table 1: Density matrices and Bloch vectors

3 Purification

3.1 Concept

Purification answers: Can a mixed state be viewed as part of a larger pure state?

Theorem 3.1 (Purification)**.** *For any density matrix ρ on system A, there exists a pure state $|\psi\rangle$ on system AR such that:*

$$\rho_A = Tr_R(|\psi\rangle\langle\psi|)$$

3.2 Physical Significance: Why Purification Matters

Key Insight: Mixedness is relative to what you have access to!

Physical Meaning:

- A mixed state on system A may actually be part of a **pure entangled state** on AR
- The "mixedness" comes from **ignoring/tracing out** the reference system R
- This is the origin of **entanglement entropy**
- In black hole physics: Hawking radiation appears mixed, but may be purified by interior
- In quantum information: Every noisy channel can be modeled as unitary on larger space

Real-world examples:

- **Thermal states**: Mixed state of system = pure state of system + heat bath
- **Decoherence**: System appears mixed because we don't track environment
- **Entanglement**: Alice's reduced state is mixed because she doesn't have Bob's qubit

3.3 Construction: Step-by-Step Proof

Proof (Constructive):

Step 1: Spectral decomposition of ρ:

$$\rho = \sum_i p_i |\psi_i\rangle\langle\psi_i|$$

Step 2: Introduce reference system R with orthonormal basis $\{|i_R\rangle\}$.
Step 3: Define pure state on AR:

$$|\psi_{AR}\rangle = \sum_i \sqrt{p_i}|\psi_i\rangle_A \otimes |i_R\rangle$$

Step 4-7: (See full derivation in text for verification)

3.4 Code: Purification Construction (Level 25)

File: `code/density_matrix/25_purification.py`

```python
def purify(rho):
    """Create purification of density matrix.

    Physical meaning:
    Any mixed state rho on A can be viewed as pure state on larger system AR.
    The mixedness comes from not having access to R.

    Construction: |psi> = sum sqrt(p_i) |psi_i>_A tensor |i>_R
    """
    # Spectral decomposition
    eigenvalues, eigenvectors = np.linalg.eigh(rho)

    # Keep only non-zero eigenvalues
    nonzero_idx = eigenvalues > 1e-10
    eigenvalues = eigenvalues[nonzero_idx]
    eigenvectors = eigenvectors[:, nonzero_idx]

    # Construct purification
    dim = len(eigenvalues)
    psi = np.zeros(dim * 2, dtype=complex)
    for i, (p, v) in enumerate(zip(eigenvalues, eigenvectors.T)):
        psi[i*2:(i+1)*2] = np.sqrt(p) * v

    return psi

# Maximally mixed state
rho_mix = np.eye(2) / 2
psi_pure = purify(rho_mix)
print("Purification of maximally mixed state:")
print(psi_pure)
print(f"Norm: {np.linalg.norm(psi_pure):.6f}")
# This is the Bell state |Phi+> = (|00> + |11>)/sqrt(2)
# Physical meaning: Maximally mixed on A = maximally entangled on AR
```

Listing 5: Create purification of density matrix (Level 25)

3.5 Example: Maximally Mixed State

For $\rho = \frac{1}{2}I$:

Purification is the Bell state:

$$|\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$$

Physical Significance:

The maximally mixed state on A arises from **maximal entanglement** with R:

- Alice has $\rho_A = I/2$ (no knowledge)
- But together, Alice + Bob share pure state $|\Phi^+\rangle$ (complete knowledge)
- Alice's ignorance is due to entanglement, not classical uncertainty
- This is the essence of "monogamy of entanglement"

4 Schmidt Decomposition

4.1 Statement

Theorem 4.1 (Schmidt Decomposition)**.** *For any pure state $|\psi\rangle$ of a composite system AB, there exist orthonormal bases $\{|i_A\rangle\}$ for A and $\{|i_B\rangle\}$ for B such that:*

$$|\psi\rangle = \sum_i \lambda_i |i_A\rangle \otimes |i_B\rangle \tag{39}$$

where $\lambda_i \geq 0$ and $\sum_i \lambda_i^2 = 1$.

Physical Significance: What Schmidt Decomposition Tells Us
Key Insights:

- **Schmidt coefficients** λ_i: Measure of entanglement
- **Schmidt rank**: Number of non-zero λ_i
- **Rank = 1**: Product state (no entanglement)
- **Rank ¿ 1**: Entangled state
- **Rank = d** (maximum): Maximally entangled (for d-dimensional systems)
- **Uniqueness**: Schmidt coefficients are unique (basis-independent entanglement measure)

Why it matters:

- **Entanglement quantification**: Schmidt coefficients determine entanglement entropy
- **Quantum protocols**: Teleportation, superdense coding rely on Schmidt structure
- **Many-body physics**: Area laws, tensor networks use Schmidt decomposition
- **Quantum chemistry**: Electronic structure calculations

4.2 Singular Value Decomposition (SVD): Complete Derivation

The Schmidt decomposition is a direct application of the Singular Value Decomposition from linear algebra. We follow the treatment in *Linear Algebra Done Right* (Axler, 3rd ed., Section 7.C).

Theorem 4.2 (Singular Value Decomposition)**.** *Let T be a linear map from finite-dimensional inner product space V to W. Then there exist orthonormal bases $\{e_1, \ldots, e_n\}$ of V and $\{f_1, \ldots, f_m\}$ of W, and nonnegative numbers $s_1, \ldots, s_n$ (called singular values) such that:*

$$Tv = s_1\langle v, e_1\rangle f_1 + \cdots + s_n\langle v, e_n\rangle f_n \tag{40}$$

for all $v \in V$.

Proof Sketch (Axler Style):
Step 1: Consider the positive operator T^*T on V. By the spectral theorem, T^*T has an orthonormal eigenbasis $\{e_1, \ldots, e_n\}$ with nonnegative eigenvalues.
Step 2: Define singular values $s_j = \sqrt{\lambda_j}$ where $T^*Te_j = \lambda_j e_j$.

Step 3: Define vectors $f_j = \frac{1}{s_j} T e_j$ for each j with $s_j > 0$. These form an orthonormal set in W (verify: $\langle f_i, f_j \rangle = \delta_{ij}$).

Step 4: Extend $\{f_j\}$ to an orthonormal basis of W.

Step 5: Conclude the SVD representation exists.

Step 6: Verify that for any $v \in V$:

$$Tv = \sum_{j=1}^{n} s_j \langle v, e_j \rangle f_j \tag{41}$$

This follows by checking that both sides have the same inner product with each f_k.

Matrix Form: In matrix notation, SVD becomes:

$$C = UDV^\dagger \tag{42}$$

where:

- U is $m \times m$ unitary (columns are left singular vectors $\{f_j\}$)
- D is $m \times n$ diagonal with singular values s_j on diagonal
- V is $n \times n$ unitary (columns are right singular vectors $\{e_j\}$)
- $V^\dagger$ denotes conjugate transpose of V

Numerical Example: Consider a non-diagonal 2×2 matrix representing a quantum state:

$$C = \begin{pmatrix} 0.5 & 0.5 \\ 0.3 & 0.7 \end{pmatrix} \tag{43}$$

This matrix is **not diagonal**, so we must compute the full SVD.

Step 1: Compute $C^\dagger C$ (for right singular vectors):

$$C^\dagger C = \begin{pmatrix} 0.5 & 0.3 \\ 0.5 & 0.7 \end{pmatrix} \begin{pmatrix} 0.5 & 0.5 \\ 0.3 & 0.7 \end{pmatrix} = \begin{pmatrix} 0.34 & 0.46 \\ 0.46 & 0.74 \end{pmatrix} \tag{44}$$

Step 2: Find eigenvalues of $C^\dagger C$ by solving characteristic equation:

$$\det(C^\dagger C - \lambda I) = \det \begin{pmatrix} 0.34 - \lambda & 0.46 \\ 0.46 & 0.74 - \lambda \end{pmatrix} = 0 \tag{45}$$

Expanding:

$$\begin{aligned} (0.34 - \lambda)(0.74 - \lambda) - (0.46)^2 &= 0 \\ \lambda^2 - 1.08\lambda + 0.2516 - 0.2116 &= 0 \\ \lambda^2 - 1.08\lambda + 0.04 &= 0 \end{aligned} \tag{46}$$

Using quadratic formula:

$$\lambda = \frac{1.08 \pm \sqrt{1.1664 - 0.16}}{2} = \frac{1.08 \pm 1.0032}{2} \tag{47}$$

This gives:

$$\lambda_1 \approx 1.0416, \quad \lambda_2 \approx 0.0384 \tag{48}$$

Step 3: Singular values are square roots of eigenvalues:

$$s_1 = \sqrt{\lambda_1} \approx 1.0206, \quad s_2 = \sqrt{\lambda_2} \approx 0.1960 \tag{49}$$

Step 4: Find eigenvectors of $C^\dagger C$ (columns of V). For $\lambda_1 \approx 1.0416$:

$$(C^\dagger C - \lambda_1 I)\vec{v}_1 = \begin{pmatrix} -0.7016 & 0.46 \\ 0.46 & -0.3016 \end{pmatrix} \vec{v}_1 = 0 \tag{50}$$

This gives $\vec{v}_1 \approx \begin{pmatrix} 0.5522 \\ 0.8337 \end{pmatrix}$ (normalized).
For $\lambda_2 \approx 0.0384$:

$$(C^\dagger C - \lambda_2 I)\vec{v}_2 = \begin{pmatrix} 0.3016 & 0.46 \\ 0.46 & 0.7016 \end{pmatrix} \vec{v}_2 = 0 \tag{51}$$

This gives $\vec{v}_2 \approx \begin{pmatrix} -0.8337 \\ 0.5522 \end{pmatrix}$ (normalized).
Thus:

$$V \approx \begin{pmatrix} 0.5522 & -0.8337 \\ 0.8337 & 0.5522 \end{pmatrix} \tag{52}$$

Step 5: Compute left singular vectors using $\vec{u}_i = \frac{1}{s_i} C\vec{v}_i$:
For $\vec{u}_1$:

$$\vec{u}_1 = \frac{1}{1.0206} \begin{pmatrix} 0.5 & 0.5 \\ 0.3 & 0.7 \end{pmatrix} \begin{pmatrix} 0.5522 \\ 0.8337 \end{pmatrix} \approx \begin{pmatrix} 0.6791 \\ 0.7341 \end{pmatrix} \tag{53}$$

For $\vec{u}_2$:

$$\vec{u}_2 = \frac{1}{0.1960} \begin{pmatrix} 0.5 & 0.5 \\ 0.3 & 0.7 \end{pmatrix} \begin{pmatrix} -0.8337 \\ 0.5522 \end{pmatrix} \approx \begin{pmatrix} -0.7341 \\ 0.6791 \end{pmatrix} \tag{54}$$

Thus:

$$U \approx \begin{pmatrix} 0.6791 & -0.7341 \\ 0.7341 & 0.6791 \end{pmatrix} \tag{55}$$

Step 6: Verify the decomposition $C = UDV^\dagger$:

$$\begin{aligned} UDV^\dagger &\approx \begin{pmatrix} 0.6791 & -0.7341 \\ 0.7341 & 0.6791 \end{pmatrix} \begin{pmatrix} 1.0206 & 0 \\ 0 & 0.1960 \end{pmatrix} \begin{pmatrix} 0.5522 & 0.8337 \\ -0.8337 & 0.5522 \end{pmatrix} \\ &\approx \begin{pmatrix} 0.5 & 0.5 \\ 0.3 & 0.7 \end{pmatrix} = C \quad \checkmark \end{aligned} \tag{56}$$

Connection to Schmidt Decomposition:
Now apply this to the quantum state:

$$|\psi\rangle = 0.5|00\rangle + 0.5|01\rangle + 0.3|10\rangle + 0.7|11\rangle \tag{57}$$

The Schmidt decomposition is:

$$|\psi\rangle = s_1|u_1\rangle \otimes |v_1\rangle + s_2|u_2\rangle \otimes |v_2\rangle \tag{58}$$

where:

$$\begin{aligned} |u_1\rangle &\approx 0.6791|0\rangle + 0.7341|1\rangle \\ |u_2\rangle &\approx -0.7341|0\rangle + 0.6791|1\rangle \\ |v_1\rangle &\approx 0.5522|0\rangle + 0.8337|1\rangle \\ |v_2\rangle &\approx -0.8337|0\rangle + 0.5522|1\rangle \end{aligned} \tag{59}$$

Physical Interpretation:

The original state looked complicated in the computational basis with four terms. But the Schmidt decomposition reveals it has only **two** correlated terms. The Schmidt coefficients $s_1 \approx 1.0206$ and $s_2 \approx 0.1960$ tell us:

- **Schmidt rank = 2**: The state is entangled
- **Not maximally entangled**: $s_1 \neq s_2$ (one term dominates)
- **Entanglement entropy**: $S = -s_1^2 \log_2(s_1^2) - s_2^2 \log_2(s_2^2) \approx 0.28$ bits

This example demonstrates that SVD works for *any* matrix, not just diagonal ones. The computational basis may hide the true entanglement structure, but Schmidt decomposition always reveals it.

Connection to Schmidt Decomposition:

For a bipartite pure state $|\psi\rangle = \sum_{i,j} c_{ij}|i\rangle_A \otimes |j\rangle_B$, form the coefficient matrix $C = (c_{ij})$. Applying SVD to C:

$$c_{ij} = \sum_k U_{ik} s_k (V^\dagger)_{kj} = \sum_k U_{ik} s_k V^*_{jk} \tag{60}$$

Substituting into the state:

$$\begin{aligned} |\psi\rangle &= \sum_{i,j} \left(\sum_k U_{ik} s_k V^*_{jk} \right) |i\rangle_A \otimes |j\rangle_B \\ &= \sum_k s_k \left(\sum_i U_{ik} |i\rangle_A \right) \otimes \left(\sum_j V^*_{jk} |j\rangle_B \right) \\ &= \sum_k s_k |k_A\rangle \otimes |k_B\rangle \end{aligned} \tag{61}$$

where we defined the Schmidt bases:

$$|k_A\rangle = \sum_i U_{ik}|i\rangle_A, \quad |k_B\rangle = \sum_j V^*_{jk}|j\rangle_B \tag{62}$$

The Schmidt coefficients are $\lambda_k = s_k$ (the singular values).

Why SVD Works for Schmidt:

The magic of SVD is that it finds the *optimal* bases for both subsystems simultaneously. In the original computational basis, the state might look complicated with many cross-terms. But SVD finds new bases where the state becomes a simple sum of product states with no cross-terms. This is analogous to how diagonalizing a matrix simplifies its action—SVD diagonalizes the *correlations* between two systems.

4.3 Proof via SVD

Proof:

Step 1: Write $|\psi\rangle$ in computational basis:

$$|\psi\rangle = \sum_{i,j} c_{ij}|i\rangle_A \otimes |j\rangle_B \tag{63}$$

Step 2: Form coefficient matrix $C = (c_{ij})$.
Step 3: Apply Singular Value Decomposition:

$$C = UDV^\dagger \tag{64}$$

Step 4: Define new bases for A and B:

$$|i_A\rangle = \sum_k U_{ki}|k\rangle_A, \quad |i_B\rangle = \sum_l V_{li}^*|l\rangle_B \tag{65}$$

Step 5: Verify orthonormality of new bases:

$$\begin{aligned}\langle i_A|j_A\rangle &= \sum_{k,l} U_{ki}^* U_{lj}\langle k|l\rangle \\ &= \sum_k U_{ki}^* U_{kj} = (U^\dagger U)_{ij} = \delta_{ij}\end{aligned} \tag{66}$$

Step 6: Substitute SVD into state expansion:

$$\begin{aligned}|\psi\rangle &= \sum_{i,j}(UDV^\dagger)_{ij}|i\rangle_A \otimes |j\rangle_B \\ &= \sum_{i,j,k} U_{ik}s_k V_{jk}^*|i\rangle_A \otimes |j\rangle_B\end{aligned} \tag{67}$$

Step 7: Rearrange summation:

$$|\psi\rangle = \sum_k s_k \left(\sum_i U_{ik}|i\rangle_A\right) \otimes \left(\sum_j V_{jk}^*|j\rangle_B\right) \tag{68}$$

Step 8: Recognize Schmidt bases from Step 4:

$$|\psi\rangle = \sum_k s_k|k_A\rangle \otimes |k_B\rangle \tag{69}$$

Step 9: Verify normalization:

$$\langle\psi|\psi\rangle = \sum_k |s_k|^2 = \mathrm{Tr}(D^2) = \mathrm{Tr}(C^\dagger C) = 1 \tag{70}$$

Step 10: Renumber steps for clarity (Step 6-10 above).
Step 6: Verify orthonormality of new bases (Eq. 66).
Step 7: Substitute SVD into state expansion (Eq. 67).
Step 8: Rearrange summation (Eq. 68).
Step 9: Recognize Schmidt bases from Step 4 (Eq. 69).
Step 10: Verify normalization (Eq. 70).

Geometric Interpretation:

The Schmidt decomposition reveals the *intrinsic geometry* of entanglement. Think of the coefficient matrix C as describing how subsystems A and B are correlated. In the computational basis, these correlations might look messy and complicated. But SVD finds the "principal axes" of correlation—directions in the Hilbert spaces of A and B where the correlations are diagonal.

This is exactly analogous to principal component analysis (PCA) in statistics: PCA finds the directions of maximum variance in data, while Schmidt decomposition finds the directions of maximum correlation between quantum systems. The Schmidt coefficients tell you how much "weight" each correlated direction carries.

Why Schmidt Rank Matters:

Schmidt rank is the number of non-zero Schmidt coefficients. This simple integer tells you everything about the entanglement structure:

- **Rank 1**: The state factors as $|\psi\rangle = |\phi_A\rangle \otimes |\phi_B\rangle$. No entanglement—A and B are independent.
- **Rank 2** (for qubits): Maximum entanglement. The state is locally equivalent to a Bell state. Measuring A instantly determines B.
- **Higher rank**: More complex entanglement structure. For d-dimensional systems, maximum rank is d.

The Schmidt rank is invariant under local unitaries—no matter how A and B rotate their local reference frames, the amount of entanglement (as measured by Schmidt rank) stays the same. This makes Schmidt rank a fundamental, basis-independent property of the state.

4.4 Code: Schmidt Decomposition (Level 26)

File: `code/density_matrix/26_schmidt_decomposition.py`

```python
def schmidt_decomposition(psi, dim_a=2, dim_b=2):
    """Compute Schmidt decomposition of bipartite state.

    Physical meaning:
    - Schmidt coefficients lambda_i: entanglement measure
    - Schmidt rank: number of non-zero coefficients
    - Rank = 1: product state (not entangled)
    - Rank > 1: entangled state

    Uses SVD: C = U D V^dagger
    Schmidt coefficients = diagonal of D
    """
    # Reshape to matrix form
    psi_matrix = psi.reshape(dim_a, dim_b)

    # SVD gives Schmidt decomposition
    U, s, Vh = np.linalg.svd(psi_matrix)

    # Schmidt coefficients
    lambdas = s

    # Schmidt bases
    basis_a = U
    basis_b = Vh.conj().T

    return lambdas, basis_a, basis_b

# Bell state |Phi+>
bell_phi_plus = np.array([1/np.sqrt(2), 0, 0, 1/np.sqrt(2)])
lambdas, ba, bb = schmidt_decomposition(bell_phi_plus)
print("Schmidt decomposition of |Phi+>:")
print("Schmidt coefficients:", lambdas)
```

```python
print("Schmidt rank:", np.sum(lambdas > 1e-10))
print("Entanglement entropy:", -np.sum(lambdas**2 * np.log2(lambdas**2)))
# Output: [0.707, 0.707], rank = 2 (maximally entangled)
# Entropy = 1 bit (maximum for 2 qubits)

# Product state |00>
product = np.array([1, 0, 0, 0])
lambdas, ba, bb = schmidt_decomposition(product)
print("\nSchmidt decomposition of |00>:")
print("Schmidt coefficients:", lambdas)
print("Schmidt rank:", np.sum(lambdas > 1e-10))
# Output: [1, 0], rank = 1 (not entangled)
# Entropy = 0 bits (no entanglement)
```

Listing 6: Compute Schmidt decomposition (Level 26)

4.5 Schmidt Rank

Definition 4.1 (Schmidt Rank)**.** *The number of non-zero Schmidt coefficients.*

- **Schmidt rank = 1:** Product state (not entangled)
- **Schmidt rank ¿ 1:** Entangled state

Physical Significance:
Schmidt rank is a **basis-independent** measure of entanglement:

- Doesn't depend on how you describe the state
- Invariant under local unitaries (local operations can't create entanglement)
- Discrete measure (unlike entanglement entropy, which is continuous)
- Determines quantum protocol capabilities (e.g., teleportation fidelity)

5 Reduced Density Matrices

5.1 Definition and Mathematical Preliminaries

Consider a bipartite quantum system AB with Hilbert space $\mathcal{H}_{AB} = \mathcal{H}_A \otimes \mathcal{H}_B$. The global state is described by a density operator ρ_{AB} acting on $\mathcal{H}_{AB}$. The **reduced density matrix** of subsystem A is defined by the **partial trace** over subsystem B:

$$\rho_A \equiv \mathrm{Tr}_B(\rho_{AB}).$$

The partial trace Tr_B is a linear map from operators on $\mathcal{H}_{AB}$ to operators on $\mathcal{H}_A$ that satisfies:

$$\mathrm{Tr}_B(X_A \otimes Y_B) = X_A \ \mathrm{Tr}(Y_B),$$

for any operators X_A on $\mathcal{H}_A$ and Y_B on $\mathcal{H}_B$, with Tr denoting the usual trace on $\mathcal{H}_B$. This definition is extended by linearity to all operators on $\mathcal{H}_{AB}$.

Key Properties of the Partial Trace

- **Preservation of Positivity:** If $\rho_{AB} \geq 0$, then $\rho_A \geq 0$.
- **Preservation of Trace:** $\mathrm{Tr}_A(\rho_A) = \mathrm{Tr}_{AB}(\rho_{AB}) = 1$.
- **Cyclic Property (with caution):** $\mathrm{Tr}_B\left((I_A \otimes M_B)\rho_{AB}\right) = \mathrm{Tr}_B\left(\rho_{AB}(I_A \otimes M_B)\right)$, but only the identity on A may be freely moved across the partial trace.
- **Reduction of Expectation Values:** For any observable O_A on A,

  $$\langle O_A \rangle_{\rho_{AB}} = \mathrm{Tr}_{AB}\left(\rho_{AB}(O_A \otimes I_B)\right) = \mathrm{Tr}_A(\rho_A O_A).$$

  Thus ρ_A encodes *all* local measurement statistics.

5.2 Physical Significance: Interpreting the Reduced Density Matrix

Central Idea: ρ_A encapsulates everything that an observer (Alice) with access **only** to subsystem A can possibly learn or predict, even if the full system AB is in a known state. The partial trace erases all information that requires access to Bob's system.

Operational and Conceptual Meanings:

- **Predictive Power:** ρ_A yields the correct expectation value $\langle O_A \rangle = \mathrm{Tr}(\rho_A O_A)$ for any observable O_A acting locally on A.
- **Ignorance Induced by the Trace:** The partial trace averages over the inaccessible degrees of freedom of B, effectively converting quantum entanglement into classical uncertainty for the local observer.
- **Entanglement Detection:** A pure global state $\rho_{AB} = |\psi\rangle\langle\psi|$ that yields a mixed reduced state ρ_A is an unmistakable fingerprint of entanglement between A and B.
- **Thermalization and Entropy:** Even when the composite system is in a pure state of zero von Neumann entropy, the reduced state can appear thermal (e.g., in eigenstate thermalization), reflecting the local observer's inability to access the full system's purity.

- **Decoherence Mechanism:** When B represents an unmonitored environment, tracing it out provides a microscopic account of how superpositions become effectively classical mixtures from the perspective of the system of interest.

Illustrative Applications:

- **Hawking Radiation:** A distant observer tracing over the interior of a black hole perceives a mixed thermal bath of radiation, even if the full quantum gravity state remains pure.
- **Open Quantum Systems Dynamics:** The evolution of a system coupled to a bath is described by a master equation derived from tracing out the environmental degrees of freedom.
- **Quantum Error Correction:** The measurement of a stabilizer syndrome is mathematically equivalent to taking a partial trace over parts of the system, collapsing the state onto a known error subspace.
- **Resource Theories of Thermodynamics:** The work extractable from a system depends intimately on its reduced state and the entanglement entropy that arises from tracing out inaccessible ancillas.

5.3 Partial Trace: Complete Explicit Derivation

We now derive the explicit formula for the partial trace in an orthonormal product basis.

Step 1: Choose Product Basis. Let $\{|i\rangle_A\}$ be an orthonormal basis for $\mathcal{H}_A$ and $\{|j\rangle_B\}$ an orthonormal basis for $\mathcal{H}_B$. Then the set $\{|i\rangle_A \otimes |j\rangle_B\}$ forms an orthonormal product basis for $\mathcal{H}_{AB}$.

Step 2: Express a General Operator in This Basis. Any operator ρ_{AB} on $\mathcal{H}_{AB}$ can be expanded as:

$$\rho_{AB} = \sum_{i,k} \sum_{j,l} \rho_{ij,kl} \; |i\rangle\langle k|_A \otimes |j\rangle\langle l|_B,$$

where $\rho_{ij,kl} = \big(\langle i|_A \otimes \langle j|_B\big)\rho_{AB}\big(|k\rangle_A \otimes |l\rangle_B\big)$.

Step 3: Apply the Partial Trace Definition. The partial trace over B is defined via its action on elementary tensor products:

$$\mathrm{Tr}_B\left(|i\rangle\langle k|_A \otimes |j\rangle\langle l|_B\right) = |i\rangle\langle k|_A \;\; \mathrm{Tr}\left(|j\rangle\langle l|_B\right).$$

Now compute the ordinary trace on B:

$$\mathrm{Tr}\left(|j\rangle\langle l|_B\right) = \sum_m \langle m|_B \left(|j\rangle\langle l|_B\right)|m\rangle_B = \sum_m \langle m|j\rangle\langle l|m\rangle = \sum_m \delta_{mj}\delta_{lm} = \delta_{jl}.$$

Thus we obtain the fundamental rule:

$$\boxed{\mathrm{Tr}_B\left(|i\rangle\langle k|_A \otimes |j\rangle\langle l|_B\right) = |i\rangle\langle k|_A \; \delta_{jl}}.$$

Step 4: Extend by Linearity. Applying this rule to the expansion of ρ_{AB} yields the explicit formula:

$$\rho_A = \mathrm{Tr}_B(\rho_{AB}) = \sum_{i,k}\sum_{j,l} \rho_{ij,kl}\ \mathrm{Tr}_B\left(|i\rangle\langle k|_A \otimes |j\rangle\langle l|_B\right) = \sum_{i,k}\left(\sum_j \rho_{ij,kj}\right)|i\rangle\langle k|_A.$$

In matrix language, if we arrange ρ_{AB} as a block matrix with blocks indexed by i, k and internal indices j, l, the reduced matrix is obtained by taking the *partial trace over the internal indices*:

$$(\rho_A)_{ik} = \sum_j \rho_{ij,kj}.$$

5.4 Example: Bell State — Complete Derivation of Reduced Density Matrix

We now carry out a fully explicit, step-by-step calculation of ρ_A for the Bell state $|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$.

Step 1: Write the Density Matrix of the Global State. The density operator for a pure state $|\psi\rangle$ is $|\psi\rangle\langle\psi|$. For $|\Phi^+\rangle$:

$$\rho_{AB} = |\Phi^+\rangle\langle\Phi^+| = \frac{1}{2}\big(|00\rangle + |11\rangle\big)\big(\langle 00| + \langle 11|\big).$$

Step 2: Expand the Outer Product. Distributing the product gives four terms:

$$\rho_{AB} = \frac{1}{2}\Big(|00\rangle\langle 00| \ + \ |00\rangle\langle 11| \ + \ |11\rangle\langle 00| \ + \ |11\rangle\langle 11|\Big).$$

Step 3: Express Each Term in Tensor Product Notation. We explicitly separate the A and B kets and bras using the tensor product symbol $\otimes$:

$$\begin{aligned}
|00\rangle\langle 00| &= (|0\rangle_A \otimes |0\rangle_B)(\langle 0|_A \otimes \langle 0|_B) = |0\rangle\langle 0|_A \otimes |0\rangle\langle 0|_B,\\
|00\rangle\langle 11| &= (|0\rangle_A \otimes |0\rangle_B)(\langle 1|_A \otimes \langle 1|_B) = |0\rangle\langle 1|_A \otimes |0\rangle\langle 1|_B,\\
|11\rangle\langle 00| &= (|1\rangle_A \otimes |1\rangle_B)(\langle 0|_A \otimes \langle 0|_B) = |1\rangle\langle 0|_A \otimes |1\rangle\langle 0|_B,\\
|11\rangle\langle 11| &= (|1\rangle_A \otimes |1\rangle_B)(\langle 1|_A \otimes \langle 1|_B) = |1\rangle\langle 1|_A \otimes |1\rangle\langle 1|_B.
\end{aligned}$$

Thus:

$$\rho_{AB} = \frac{1}{2}\Big(|0\rangle\langle 0|_A \otimes |0\rangle\langle 0|_B + |0\rangle\langle 1|_A \otimes |0\rangle\langle 1|_B + |1\rangle\langle 0|_A \otimes |1\rangle\langle 0|_B + |1\rangle\langle 1|_A \otimes |1\rangle\langle 1|_B\Big).$$

Step 4: Apply the Partial Trace Rule to Each Term. Using $\mathrm{Tr}_B(|i\rangle\langle k|_A \otimes |j\rangle\langle l|_B) = |i\rangle\langle k|_A \cdot \delta_{jl}$, we compute:

$$\begin{aligned}
\mathrm{Tr}_B\big(|0\rangle\langle 0|_A \otimes |0\rangle\langle 0|_B\big) &= |0\rangle\langle 0|_A \cdot \delta_{00} = |0\rangle\langle 0|_A,\\
\mathrm{Tr}_B\big(|0\rangle\langle 1|_A \otimes |0\rangle\langle 1|_B\big) &= |0\rangle\langle 1|_A \cdot \delta_{01} = 0,\\
\mathrm{Tr}_B\big(|1\rangle\langle 0|_A \otimes |1\rangle\langle 0|_B\big) &= |1\rangle\langle 0|_A \cdot \delta_{10} = 0,\\
\mathrm{Tr}_B\big(|1\rangle\langle 1|_A \otimes |1\rangle\langle 1|_B\big) &= |1\rangle\langle 1|_A \cdot \delta_{11} = |1\rangle\langle 1|_A.
\end{aligned}$$

Step 5: Sum the Surviving Terms. Only the diagonal terms (with $j = l$) survive:

$$\rho_A = \mathrm{Tr}_B(\rho_{AB}) = \frac{1}{2}\big(|0\rangle\langle 0|_A + |1\rangle\langle 1|_A\big).$$

Step 6: Identify the Result. The sum $|0\rangle\langle 0| + |1\rangle\langle 1|$ is the identity operator I_A on the two-dimensional Hilbert space of A. Hence:

$$\rho_A = \frac{1}{2} I_A = \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{2} \end{pmatrix}.$$

Step 7: Verify Properties.

- **Trace:** $\mathrm{Tr}(\rho_A) = \frac{1}{2} + \frac{1}{2} = 1$ (correct normalization).
- **Purity:** $\mathrm{Tr}(\rho_A^2) = \mathrm{Tr}\big(\frac{1}{4} I\big) = \frac{1}{2} < 1$, so ρ_A is mixed.
- **Entropy:** The von Neumann entropy is $S(\rho_A) = -\mathrm{Tr}(\rho_A \log_2 \rho_A) = 1$ bit, the maximum for a qubit.

5.5 Physical Consequences and the Entanglement Paradox

The result $\rho_A = I/2$ embodies the profound counterintuitiveness of quantum correlations:

- **Global Purity:** The state $|\Phi^+\rangle$ represents maximal knowledge about the composite system AB — its von Neumann entropy is zero.
- **Local Ignorance:** The reduced matrix ρ_A is maximally mixed, signifying that Alice, acting in isolation, possesses **zero** information about the outcome of any measurement she might perform.
- **Random Yet Correlated:** Alice observes a perfectly random sequence of results (equal probability for $|0\rangle$ and $|1\rangle$). However, every one of her outcomes is perfectly synchronized with a corresponding outcome on Bob's side.
- **Where Did the Information Go?** The missing information about the state of A is not destroyed; it is entirely relocated to the non-local correlations between A and B. The joint entropy vanishes, while the marginal entropy of each subsystem is maximal.

Essential Takeaway: A pure global state that reduces to a mixed local state is the quintessential signature of bipartite entanglement. The information is not lost—it resides in the invisible web of correlations that cannot be captured by looking at either half of the system alone.

5.6 Connection to Schmidt Decomposition and Entanglement Measures

For any pure bipartite state $|\psi\rangle_{AB}$, the Schmidt decomposition provides a direct way to compute the reduced density matrices. Writing

$$|\psi\rangle_{AB} = \sum_{i=1}^{r} \sqrt{\lambda_i}\, |u_i\rangle_A \otimes |v_i\rangle_B,$$

where $\lambda_i > 0$ are the Schmidt coefficients (with $\sum_i \lambda_i = 1$) and r is the Schmidt rank, the partial traces yield:

$$\rho_A = \sum_{i=1}^{r} \lambda_i \, |u_i\rangle\langle u_i|_A, \qquad \rho_B = \sum_{i=1}^{r} \lambda_i \, |v_i\rangle\langle v_i|_B.$$

Thus ρ_A and ρ_B share the same non-zero eigenvalues $\{\lambda_i\}$. The amount of entanglement is quantified by the von Neumann entropy of either reduced state:

$$E(|\psi\rangle_{AB}) = S(\rho_A) = -\sum_i \lambda_i \log_2 \lambda_i.$$

For the Bell state, $\lambda_1 = \lambda_2 = 1/2$, giving maximal entanglement entropy of 1 bit. This formalism clarifies that the partial trace not only discards Bob's degrees of freedom but also reveals the spectrum of the quantum correlations linking Alice and Bob.

5.7 Code: Reduced Density Matrix (Level 27)

File: `code/density_matrix/27_reduced_density_matrix.py`

```python
def reduced_density_matrix(rho_ab, keep='A', dim=2):
    """Trace out one subsystem from density matrix.

    Physical meaning:
    rho_A = Tr_B(rho_AB) describes what Alice can measure
    without access to Bob. Information about correlations
    is lost, but local predictions are complete.
    """
    # Reshape to tensor form (dim x dim x dim x dim)
    rho_tensor = rho_ab.reshape(dim, dim, dim, dim)

    if keep == 'A':
        # Trace out B (axes 1 and 3)
        rho_a = np.trace(rho_tensor, axis1=1, axis2=3)
    else:
        # Trace out A (axes 0 and 2)
        rho_a = np.trace(rho_tensor, axis1=0, axis2=2)

    return rho_a

# Bell state |Phi+>
bell_phi_plus = np.array([1/np.sqrt(2), 0, 0, 1/np.sqrt(2)])
rho_bell = np.outer(bell_phi_plus, bell_phi_plus.conj())
print("Bell state density matrix:")
print(rho_bell)

# Trace out B
rho_a = reduced_density_matrix(rho_bell, keep='A')
print("\nReduced density matrix (trace out B):")
print(rho_a)
print("\nResult: Maximally mixed state I/2")
print("Physical meaning: Pure global state -> Mixed local state")
print("This is the signature of entanglement!")
print("Alice sees randomness, but outcomes correlate with Bob's")
```

Listing 7: Compute reduced density matrix (Level 27)

5.8 Entanglement Criterion: Proof

Theorem 5.1. *A pure state $|\psi\rangle_{AB}$ is entangled if and only if its reduced density matrix ρ_A is mixed.*

Physical Significance:
This theorem gives an **operational test** for entanglement:

- Compute $\rho_A = \mathrm{Tr}_B(|\psi\rangle\langle\psi|)$
- Check if $\mathrm{Tr}(\rho_A^2) < 1$ (mixed)
- If yes: state is entangled
- If no: state is product state

Proof: (See Section 5.5 for complete bidirectional proof)

6 Entanglement Detection

6.1 Code: Entanglement Check (Level 28)

File: `code/density_matrix/28_entanglement_check.py`

```python
def schmidt_decomposition(psi, dim_a=2, dim_b=2):
    """Compute Schmidt decomposition."""
    psi_matrix = psi.reshape(dim_a, dim_b)
    U, s, Vh = np.linalg.svd(psi_matrix)
    return s, U, Vh.conj().T

def is_entangled(psi, dim_a=2, dim_b=2):
    """Check if pure state is entangled using Schmidt rank.

    Physical meaning:
    - Schmidt rank = 1: Product state (no entanglement)
    - Schmidt rank > 1: Entangled state
    - Higher rank = more entanglement

    This is a basis-independent, operational test.
    """
    lambdas, _, _ = schmidt_decomposition(psi, dim_a, dim_b)
    schmidt_rank = np.sum(lambdas > 1e-10)
    return schmidt_rank > 1

# Test product states
print("Product states (not entangled):")
print("|00> entangled:", is_entangled(np.array([1, 0, 0, 0])))
print("|01> entangled:", is_entangled(np.array([0, 1, 0, 0])))
print("|10> entangled:", is_entangled(np.array([0, 0, 1, 0])))
print("|11> entangled:", is_entangled(np.array([0, 0, 0, 1])))

# Test Bell states
print("\nBell states (maximally entangled):")
bell_phi_plus = np.array([1/np.sqrt(2), 0, 0, 1/np.sqrt(2)])
bell_phi_minus = np.array([1/np.sqrt(2), 0, 0, -1/np.sqrt(2)])
bell_psi_plus = np.array([0, 1/np.sqrt(2), 1/np.sqrt(2), 0])
bell_psi_minus = np.array([0, 1/np.sqrt(2), -1/np.sqrt(2), 0])
print("|Phi+> entangled:", is_entangled(bell_phi_plus))
print("|Phi-> entangled:", is_entangled(bell_phi_minus))
print("|Psi+> entangled:", is_entangled(bell_psi_plus))
print("|Psi-> entangled:", is_entangled(bell_psi_minus))

# Test partially entangled state
partial = np.array([0.6, 0, 0, 0.8])
print("\nPartially entangled (0.6|00> + 0.8|11>):")
print("Entangled:", is_entangled(partial))
print("Note: Schmidt rank = 2, but not maximal entanglement")
```

Listing 8: Check if state is entangled (Level 28)

7 Complete Analysis Tool

7.1 Code: Density Matrix Analyzer (Level 29)

File: `code/density_matrix/29_complete_analyzer.py`

```python
class DensityMatrixAnalyzer:
    """Complete analysis of density matrices.

    Physical quantities computed:
    - Purity: Tr(rho^2), measures how close to pure state
    - Entropy: -Tr(rho log_2 rho), measures uncertainty/entanglement
    - Bloch vector: geometric representation
    - Eigenvalues: physical measurement probabilities
    """

    def __init__(self, rho):
        self.rho = rho
        self.valid, self.msg = self._is_valid(rho)

    def _is_valid(self, rho, tol=1e-10):
        """Check if matrix is valid density matrix."""
        if not np.allclose(rho, rho.conj().T):
            return False, "Not Hermitian"
        if not np.abs(np.trace(rho) - 1) < tol:
            return False, "Trace != 1"
        eigenvalues = np.linalg.eigvalsh(rho)
        if np.any(eigenvalues < -tol):
            return False, "Not PSD"
        return True, "Valid"

    def eigenvalues(self):
        """Get eigenvalues. Physical meaning: measurement probabilities."""
        return np.linalg.eigvalsh(self.rho)

    def von_neumann_entropy(self):
        """S(rho) = -Tr(rho log_2 rho).

        Physical meaning:
        - For pure states: S = 0 (no uncertainty)
        - For mixed states: S > 0 (classical + quantum uncertainty)
        - For maximally mixed: S = log_2(d) (maximum uncertainty)
        - For entangled pure state: S_A = entanglement entropy
        """
        eigenvalues = self.eigenvalues()
        eigenvalues = eigenvalues[eigenvalues > 1e-10]
        return -np.sum(eigenvalues * np.log2(eigenvalues))

    def purity(self):
        """Pur(rho) = Tr(rho^2).

        Physical meaning:
        - Pure states: Pur = 1 (complete knowledge)
        - Mixed states: Pur < 1 (partial knowledge)
        - Maximally mixed: Pur = 1/d (minimum knowledge)
        """
        return np.trace(self.rho @ self.rho).real

    def bloch_vector(self):
```

```python
        """Extract Bloch vector.

        Physical meaning:
        - Direction: spin preference axis
        - Length: purity (1=pure, 0=maximally mixed)
        """
        X = np.array([[0, 1], [1, 0]])
        Y = np.array([[0, -1j], [1j, 0]])
        Z = np.array([[1, 0], [0, -1]])
        return np.array([
            np.trace(self.rho @ X).real,
            np.trace(self.rho @ Y).real,
            np.trace(self.rho @ Z).real
        ])

    def report(self):
        """Print complete analysis report."""
        print(f"Valid: {self.valid} ({self.msg})")
        print(f"Eigenvalues: {self.eigenvalues()}")
        print(f"Purity: {self.purity():.4f}")
        print(f"Entropy: {self.von_neumann_entropy():.4f}")
        print(f"Bloch vector: {self.bloch_vector()}")
        if np.isclose(self.purity(), 1.0):
            print("Interpretation: Pure state (complete knowledge)")
        else:
            print("Interpretation: Mixed state (partial knowledge)")

# Test
psi_plus = np.array([1/np.sqrt(2), 1/np.sqrt(2)])
rho_plus = np.outer(psi_plus, psi_plus.conj())
print("=== Pure State |+> ===")
DensityMatrixAnalyzer(rho_plus).report()

print("\n=== Maximally Mixed State ===")
DensityMatrixAnalyzer(np.eye(2)/2).report()

print("\n=== Partially Mixed State ===")
DensityMatrixAnalyzer(np.array([[0.7, 0], [0, 0.3]])).report()
```

Listing 9: Complete density matrix analyzer (Level 29)

8 Summary & Exercises

8.1 Key Concepts

- **Density matrix:** $\rho = \sum_i p_i |\psi_i\rangle\langle\psi_i|$ – encodes both quantum and classical uncertainty
- **Pure state:** $\rho = |\psi\rangle\langle\psi|$, $\mathrm{Tr}(\rho^2) = 1$ – complete knowledge
- **Mixed state:** $\mathrm{Tr}(\rho^2) < 1$ – partial knowledge
- **Bloch vector:** $\rho = \frac{1}{2}(I + \vec{r} \cdot \vec{\sigma})$, $|\vec{r}| \leq 1$ – geometric representation
- **Purification:** Mixed state as part of larger pure state – entanglement with environment
- **Schmidt decomposition:** $|\psi\rangle = \sum_i \lambda_i |i_A\rangle \otimes |i_B\rangle$ – canonical entanglement form
- **Schmidt rank:** Entanglement measure for pure states – basis-independent
- **Reduced density matrix:** $\rho_A = \mathrm{Tr}_B(\rho_{AB})$ – local description
- **Entanglement criterion:** Pure global $\rightarrow$ mixed local $\Leftrightarrow$ entangled

8.2 Physical Significance Summary

- **Density matrices** unify quantum superposition and classical probability
- **Coherences** (off-diagonal) enable interference – destroyed by decoherence
- **Mixedness** can arise from entanglement (quantum) or ignorance (classical)
- **Purification** shows mixedness is relative to accessible systems
- **Schmidt decomposition** reveals intrinsic entanglement structure
- **Reduced density matrices** describe local observations in entangled systems
- **Entanglement entropy** quantifies quantum correlations

8.3 Exercises

1. Derive the density matrix for $|-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}$ step by step.
2. Prove that $\mathrm{Tr}(\rho^2) = \frac{1+|\vec{r}|^2}{2}$ for any single-qubit state.
3. Compute purification of $\rho = 0.7|0\rangle\langle 0| + 0.3|1\rangle\langle 1|$ and verify.
4. Find Schmidt decomposition of $|\psi\rangle = 0.6|00\rangle + 0.8|11\rangle$ using SVD.
5. Show that reduced density matrix of Bell state is maximally mixed (complete calculation).
6. Compute von Neumann entropy for $\rho = p|0\rangle\langle 0| + (1-p)|1\rangle\langle 1|$ as function of p.
7. Prove that local unitaries cannot change Schmidt coefficients.
8. **Physical interpretation:** Explain why a pure entangled state gives mixed reduced states.
9. **Thermal states:** Show that $\rho = e^{-\beta H}/Z$ is mixed at finite temperature.
10. **Decoherence:** Model dephasing as decay of off-diagonal elements.

8.4 Milestones

- ☐ Create density matrices from state vectors
- ☐ Distinguish pure vs mixed states (purity test)
- ☐ Extract Bloch vectors from density matrices
- ☐ Construct purifications (spectral decomposition)
- ☐ Compute Schmidt decompositions (via SVD)
- ☐ Calculate reduced density matrices (partial trace)
- ☐ Detect entanglement via Schmidt rank
- ☐ Compute von Neumann entropy
- ☐ Interpret physical meaning of all quantities